

Phase 3 Data-Quality Audit

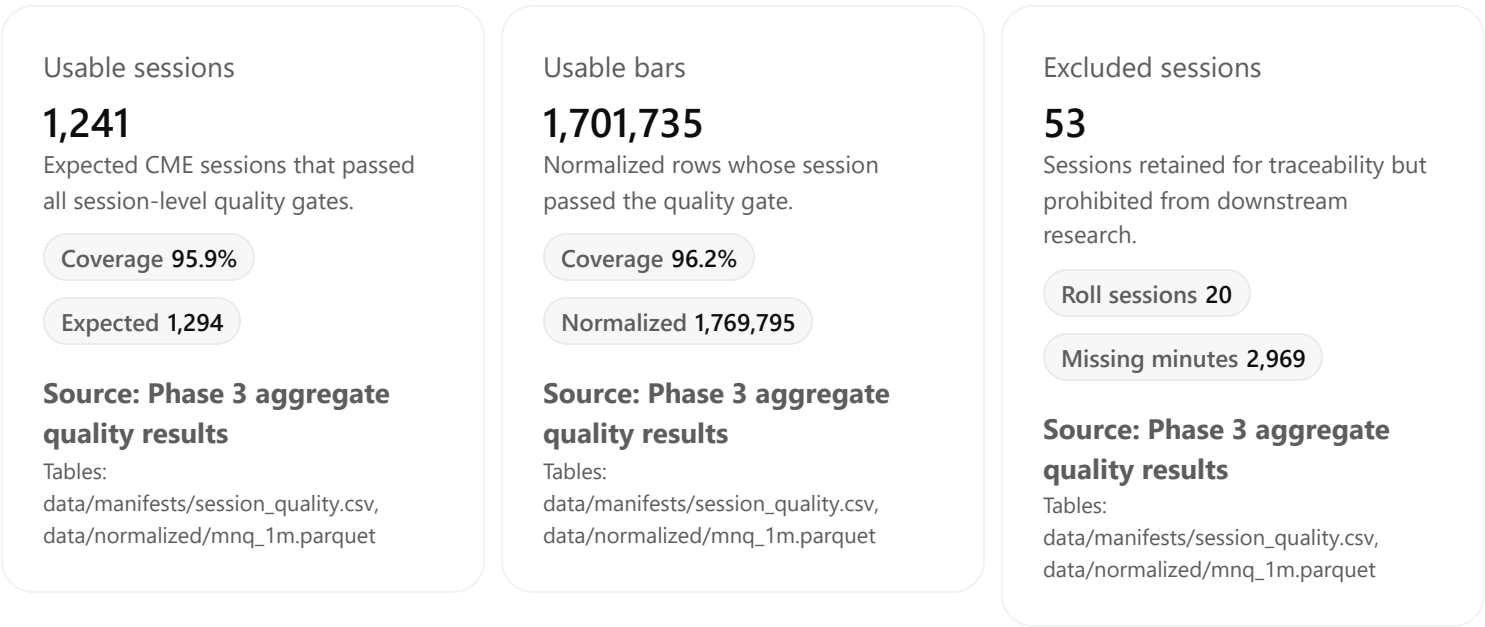
Technical audit of the normalized MNQ one-minute discovery dataset.

Technical summary

The Phase 3 gate passed. The normalized MNQ dataset is suitable for feature construction only when downstream code filters to `session_usable = true`. The audit retained **1,241 of 1,294 expected sessions (95.9%)** and **1,701,735 of 1,769,795 normalized bars (96.2%)**. All 265 raw responses passed checksum verification, with no invalid OHLCV rows or duplicate keys.

Source: Phase 3 aggregate quality results

Tables: data/manifests/session_quality.csv, data/normalized/mnq_1m.parquet



Usable coverage is consistently high, with explicit exclusions every year

Usable sessions exceed excluded sessions by a wide margin in every calendar year. The chart keeps exclusions visible instead of hiding them through imputation. The 53 excluded sessions consist of 20 mixed-contract roll sessions plus conservative exclusions for provider gaps or timestamps outside the expected interval.

Source: Phase 3 session-quality records

Table: data/manifests/session_quality.csv

Session quality by calendar year

Source: Phase 3 session-quality records

Table: data/manifests/session_quality.csv

Year	Session Count	Status
2.02K	106	Usable
2.02K	6	Excluded
2.02K	249	Usable
2.02K	9	Excluded
2.02K	248	Usable
2.02K	10	Excluded
2.02K	249	Usable
2.02K	11	Excluded
2.03K	248	Usable
2.03K	11	Excluded
2.03K	141	Usable
2.03K	6	Excluded

The locked final-test boundary remains intact

Development retains 744 usable sessions, validation retains 250, and the final untouched test retains 247. Excluded sessions stay inside their original chronological split; the boundaries were not moved in response to prices, returns, or strategy results.

Source: Phase 3 session-quality records

Table: data/manifests/session_quality.csv

Session quality by chronological split

Source: Phase 3 session-quality records

Table: data/manifests/session_quality.csv

Split	Quality records	Usable sessions	Excluded sessions	Usable rate
Development	777	744	33	95.8%
Final untouched test	259	247	12	95.4%
Validation	258	250	8	96.9%

Scope and definitions

The source population is 265 immutable one-minute MNQ responses covering trade dates July 28, 2021 through July 27, 2026. A CME Globex Equity session normally runs from 18:00 Eastern on the prior calendar day through 17:00 Eastern on the trade date. Bar timestamps denote the start of a half-open one-minute interval. RTH is 09:30-15:59 Eastern.

Deterministic normalization and audit method

Each raw SHA-256 was recalculated before parsing. Bars were validated for schema, numeric finiteness, OHLC ordering, positive prices, non-negative volume and ticks, minute alignment, and 0.25-point tick alignment. UTC was preserved, Eastern timestamps include their DST offset, and every observed minute was reconciled to the product calendar. Exact duplicates would retain the lexicographically first request and source row; none were found.

Limitations and robustness boundaries

NQ confirmation remains deferred under the Phase 2 scope. Two Good Friday sessions have no provider bars relative to the product calendar and are excluded without imputation. All roll-date sessions are excluded because Phase 2 changed contracts at midnight Eastern rather than the session open. The final-test interval was audited only for structural quality - no signals or performance were examined.

Phase 4 requirements

1. Filter all research inputs to `session_usable = true`.
2. Keep the finalized split labels and locked final-test boundary unchanged.
3. Make every feature's availability timestamp explicit.
4. Invalidate prior-session and overnight features whenever a prerequisite session is excluded.
5. Prevent feature-analysis output from consulting the final-test split.

Further questions

Session-aligned roll data can be reacquired later if a frozen finalist depends on roll weeks. Likewise, missing non-roll sessions can be re-requested before granular finalist validation without changing the chronological boundaries.